

Cyber Lab Module 4 – Enumeration & Hydra Fundamentals

Objective

Learn the workflow for basic network enumeration in a controlled lab using Kali Linux and Metasploitable.

Lab Topology (Sanitized)

Attacker (Kali): 192.168.x.x

Target (Metasploitable): 192.168.x.x

Workflow

1. Verify connectivity: `ping 192.168.x.x`

Purpose: Confirm the target is reachable.

2. Discover open ports: `nmap 192.168.x.x`

Purpose: Identify listening services. Key findings included FTP (21), SSH (22), SMB (445), PostgreSQL (5432).

3. Enumerate SMB users:

`nmap --script=smb-enum-users.nse -p 445 192.168.x.x`

Purpose: Collect account information exposed by SMB on an intentionally vulnerable training system.

4. Detect PostgreSQL version:

`nmap -sV -p 5432 192.168.x.x`

Purpose: Identify the service and version running on the database port.

Hydra Fundamentals

Hydra is a password auditing tool used in authorized environments.

-L = username list file.

-P = password list file.

Credentials = Username + Password.

Wordlist = Text file containing candidate usernames or passwords.

Key Terms

Enumeration: Gathering information about systems before deeper testing.

SMB: Windows file sharing protocol (port 445).

PostgreSQL: Database service (default port 5432).

-sV: Detect service version.

Learning Outcome

This module demonstrated a professional assessment workflow: Reachability → Port Scan → Enumeration → Service Detection → Authentication Assessment Planning.